LEVELHULL · DESIGN DECISION RECORD

LevelHull design for construction

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DOCUMENT CONTROL

Document ID	LVH-DDR-002	Version	0.2
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/decisions/0002-design-for-construction.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	C3 cross-referenced to LVH-DDR-003 (modules widened to 220 mm)
0.1	2026-10-03	Amish Chadha	Design made constructable; changes C1 to C12 and assumptions A1 to A5 decided under Amish's 2026-10-03 pre-approval

0002: Design for construction

- **Date:** 2026-10-03
- **Status:** decided. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Context

STANDARDS section 18 asks that every part can be made by a stated process and fits and fastens to its neighbours. The TRL 1 concept named "foam blocks in protective covers, strapped and fixed" with "fixing straps and cleats", but said nothing about what the straps pull against, how a block sits on a flared side, or where the lift goes into the hull. The constructability review of the model (`cad/src/model.py` , 126 checks: overlaps with planks, frames and thwarts; contact between module, batten and frames; screw depth in the frames; straps clear of the frames; eye bolts through the strake; gauge fit) led to the changes below. None changes what the product does or its pitch. C1 follows D1 of LVH-DDR-001.

Changes

Table 1. Changes made for construction.

#	The concept had	The constructable design has	Why
C1	Blocks low along the inside faces	Modules high under the gunwale, top 30 mm below it, outboard face on the frames' inner faces	Roll stability when swamped (D1); the frame faces form a flat, regular bearing surface
C2	One block per side, size open	Four modules a side between the thwarts: 880, 1,400, 1,400 and 1,100 mm	Thwarts stay in place; each module is light (under 4 kg) and handled by one person
C3	Rectangular blocks	Parallelogram section 200 x 200 mm (widened to 220 mm by LVH-DDR-003): four equal 50 mm strips with both long edges cut at the flare angle, each layer set 31 mm further out	Sits flat on the sloping frame faces; every strip is the same cut
C4	"Cleats"	A hardwood shelf batten under each module, 50 mm thick square to the frame faces, 70 mm high, 40 mm past each module end	Gives the module a seat and the straps an anchor, and spreads the lift over two or three frames
C5	Fixings unspecified	One M8 x 100 coach screw into every frame a batten crosses, square to the batten face, 50 mm into the frame, tip 10 mm short of the planks	No new holes in the planking below the waterline; screw factor 5.5 on the full lift
C6	Nothing stopping fore and aft movement	End chocks 40 mm long and 50 mm high screwed to the batten at both ends of each module	The straps alone could let a module creep along the batten
C7	"Fixing straps"	Closed loops of 50 mm polyester webbing round module and batten, two per module, in the bays between frames, running outboard of the module in the 60 mm gap at the planks; stainless double D-ring friction buckles	A loop needs no anchor of its own; D-rings have no spring or cam to corrode
C8	Foam blocks	Layers held by the cover and straps, no glue	Polyethylene foam does not glue reliably with common adhesives
C9	Protective covers	Sewn PVC-coated polyester tarpaulin sleeves, 150 mm end flaps laced through eyelets	A tailor or sail maker can make and replace them; foam can be inspected
C10	Hull sizes taken by hand	A plywood setting gauge hooked over the gunwale marks the batten top and bottom on every frame	Every batten at the same height on a hull without drawings
C11	Nothing to hold	Four M8 eye bolts a side through the rubbing strake and top plank, between frames, and a 12 mm grab line in bights	D7; above every waterline
C12	"Freeboard mark", "bailer"	A painted band whose top edge is 50 mm below the gunwale amidships, both sides; two 10 L buckets on lanyards stowed between frames	The R2 check is visible in the swamp test; bailers cannot float away

Assumptions decided with the changes

- **A1.** The canoe's frames are at most 600 mm apart and at least 60 mm deep from the planks, so each batten crosses two frames and each screw has 50 mm of wood. Conservative: on a hull with shallower frames the screw is shortened to keep 10 mm clear of the planks and a third frame is added to the batten; never through the planking.

- **A2.** Wet hull timber is as dense as water (D3). Relaxed only by weighing and measuring the hull.
- **A3.** Frame-to-plank fastenings are sound. Each frame takes at most 0.19 kN of lift; fastenings are inspected before fitting and any loose or rotten frame is repaired first.
- **A4.** Foam density 30 kg/m3 and water uptake under 1 % by volume, to confirm from the supplier's data sheet and by immersion at TRL 4.
- **A5.** Strengths of the webbing (10 kN), D-ring pair (4 kN) and coach screw (1.5 kN sideways) to confirm with the parts bought.

Consequences

- The kit has 13 kinds of part, five of them made (gauge, battens, chocks, cores, covers) with saws, a plane, a drill, a knife and a sewing machine.
- Kit mass 67 kg; the battens are the heaviest part (39 kg with the chocks).
- The general arrangement LVH-DWG-001 is at Rev P2; making sketches LVH-DWG-101 to 106; build plan LVH-BLD-001.
- `design_state: constructable` in `project.yaml`.

> **Safety:** The fixings carry the full lift of a module pushed under water. Every batten is pull-tested by hand and one bay per canoe is proof-loaded to twice its share before the swamp test (LVH-BLD-001, sections 5 and 6).